

Lecture 3

Superposition and quantum interference

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Information is physical

Our previous lecture discussed different models of information and how to represent them in a computational device:

Information

- **Deterministic**, e.g. $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$
- **Probabilistic**, e.g. $\begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$
- **Quantum**, e.g. $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1+i}{2} \end{bmatrix}$

Some sort of vectors, over different fields, seem to provide not so different formal models for each kind of information... **However**,

Quantum information is a different story

A quantum state holds the information of **both** possible classical states:



A unit of information lives in a 2-dimensional **complex** vector space:

$$|\nu\rangle = \alpha|0\rangle + \beta|1\rangle$$

and thus possesses a **continuum of possible values**, so potentially, can store lots of classical data.

Quantum information is a different story

In a sense, all this potential is **hidden**:

when **observed** $|v\rangle$ **collapses into a classic state**: $|0\rangle$, with probability $|\alpha|^2$,
or $|1\rangle$, with probability $|\beta|^2$.

(Recall: $|\alpha| = \sqrt{\alpha\bar{\alpha}}$ for a complex α)

The outcome of an observation is **probabilistic**, which calls for a restriction to **unit** vectors, i.e. st

$$|\alpha|^2 + |\beta|^2 = 1$$

to represent quantum states.

Quantum information is a different story

This quantum state is **not** a probabilistic mixture: it is **not** true that the state is really either $|0\rangle$ or $|1\rangle$ and we just do not happen to know which.

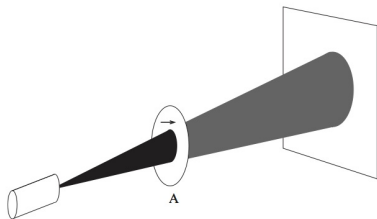
Coefficients in the linear combination are not real numbers (e.g. probabilities) that can only increase when added.

In a quantum state, such coefficients are called **amplitudes**: they are **complex** numbers, and can therefore **cancel each other or lower the probability** they entail, thus capturing a fundamental **quantum resource**:

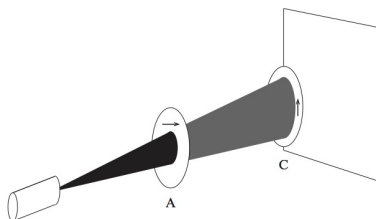
interference

Do they model some physical phenomena detected in Nature?

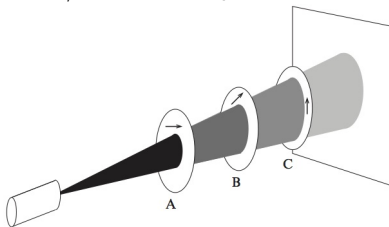
Quantum information: An experiment with a photon



$|0\rangle$ - horizontal polarization



$|1\rangle$ - vertical polarization



(from [Reifell & Polak, 2011])

Quantum information: An experiment with a photon

An explanation for a *single* photon experiment

- The photon's polarization state is modelled by a unit vector, for example, the following **linear combination of $|0\rangle$ and $|1\rangle$** :

$$\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

which corresponds to a polarization of 45 degrees.

- On passing a polaroid the photon will be absorbed or leave the polaroid with its polarization aligned with the polaroid's axis.
- The probability to go through the polaroid is the **square of the magnitude** of the amplitude of its polarization in the direction of the polaroid's axis.

Quantum information: An experiment with a photon

- Polarization of polaroid B is $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$
- This vector, together with $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$ forms a **basis** for the 2-dimensional vector space
- of which $\{|0\rangle, |1\rangle\}$ is another one (the so-called **computational basis**).
- Expressing $|0\rangle$ in terms of $|+\rangle$ and $|-\rangle$ yields

$$|0\rangle = \frac{1}{\sqrt{2}}|-\rangle + \frac{1}{\sqrt{2}}|+\rangle$$

which explains why a visible effect appears in the wall:

the photon goes through C with 50% of probability
(i.e. $|\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$).

Why aren't probabilities enough?

Computation is always a physical process

That's our *motto*!

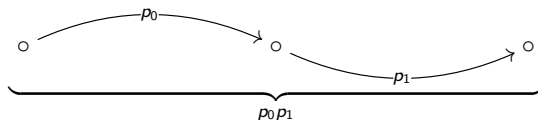
In several cases, the language of **probability theory** can describe the actual **physical evolution of a system**, i.e.

- **Physics** identify the system's structure and assign numerical probabilities to elementary transition steps.
- **Probability theory**, i.e., the Kolmogorov axioms, ensure mathematical consistency and help in calculating probabilities along paths of evolution.

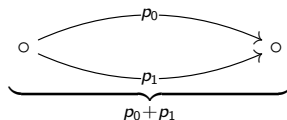
but **not always**!

Probabilistic systems, probabilistic computation

Sequential paths



Alternative, mutually exclusive paths



(Kolmogorov (1903-1987) [additivity axiom](#))

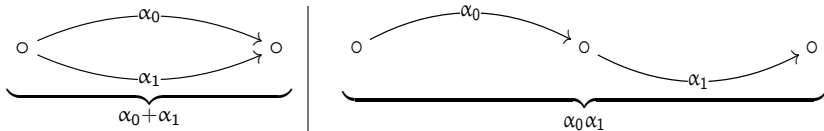
Quantum systems, quantum computation

Many common quantum phenomena, however, cannot be described this way, but are accommodated by a **modified 'probability theory'**:

Transitions are labelled by **complex** numbers, called their **amplitudes**, whose **norms squared** are interpreted as **transition probabilities** through

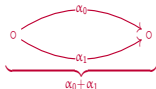
$$\text{Born's rule } p = |\alpha|^2$$

(for Max Born, 1882-1970)



Quantum systems, quantum computation

Let's compute the total probability in



$$\begin{aligned} p &= |\alpha_0 + \alpha_1|^2 \\ &= \overline{(\alpha_0 + \alpha_1)}(\alpha_0 + \alpha_1) \\ &= (\overline{\alpha_0} + \overline{\alpha_1})(\alpha_0 + \alpha_1) \\ &= |\alpha_0|^2 + |\alpha_1|^2 + \overline{\alpha_0}\alpha_1 + \alpha_0\overline{\alpha_1} \\ &= p_0 + p_1 + |\alpha_0||\alpha_1| \left(e^{i(\varphi_1 - \varphi_0)} + e^{-i(\varphi_1 - \varphi_0)} \right) \\ &= p_0 + p_1 + \underbrace{2\sqrt{p_0 p_1} \cos(\varphi_1 - \varphi_0)}_{\text{interference}} \end{aligned}$$

(expressing α_j in polar form $|\alpha_j| e^{i\theta_j}$
and resorting to $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$)

Quantum systems, quantum computation

$$p = p_0 + p_1 + \underbrace{2\sqrt{p_0 p_1} \cos(\varphi_1 - \varphi_0)}_{\text{interference}}$$

- The total probability is the sum of the probabilities of the individual transitions **modified by the interference term**.
- Depending on term $\varphi_1 - \varphi_0$ the interference can be either **negative** or **positive**.
- The important quantity is the **relative phase $\varphi_1 - \varphi_0$** rather than individual φ_0, φ_1 .
- If the system's evolution depends only on that difference, then the system **must have, somehow, experienced both paths**.

Quantum systems, quantum computation

Indeed, the **probabilistic** assumption that one or the other transition occurs, or that the system presents one or the other configuration, but **we just do not know which one**, is inconsistent with many experiments:

Typically, a quantum state is a **superposition** of basic states, i.e. states that, mathematically, form a **basis** of the vector space in which such states live — e.g. $\{|i\rangle \mid i = 1 \cdots n\}$

$$|\varphi\rangle = \sum_i \alpha_i |i\rangle$$

Recall



Superposition in action

A random number generator

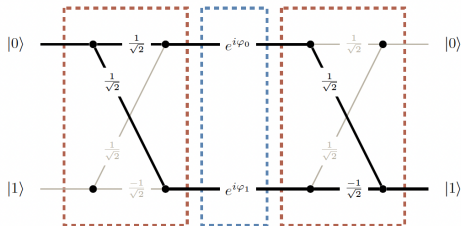
- Prepare quantum state

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

- Measure $|+\rangle$ in the computational basis to obtain either 0 or 1 with equal probability ($|\frac{1}{\sqrt{2}}|^2 = 0.5$)

This algorithm produces a perfect random number even though no randomness has been used inside it.

Ramsey interference experiment



(from [Ekert *et al*, 2024])

Atoms are sent through two separate resonant interaction zones (which attempt to commute the atom between ground and excited states), separated by an intermediate dispersive interaction zone (which performs a phase shift).

The atoms are subsequently measured and found to be in one of the two basic energy states labeled as $|0\rangle$ and $|1\rangle$.

What is the probability of an atom going from $|0\rangle$ to $|1\rangle$?

First compute the **amplitudes**:

$$\begin{aligned}A_{10} &= \frac{1}{\sqrt{2}} e^{i\varphi_0} \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} e^{i\varphi_1} \frac{-1}{\sqrt{2}} \\&= \frac{1}{2} (e^{i\varphi_0} - e^{i\varphi_1}) \\&= \frac{1}{2} \left(e^{i\frac{\varphi_0+\varphi_1}{2}} e^{i\frac{\varphi_0-\varphi_1}{2}} - e^{i\frac{\varphi_0+\varphi_1}{2}} e^{-i\frac{\varphi_0-\varphi_1}{2}} \right) \\&= \frac{1}{2} e^{i\frac{\varphi_0+\varphi_1}{2}} \left(e^{i\frac{\varphi_0-\varphi_1}{2}} - e^{-i\frac{\varphi_0-\varphi_1}{2}} \right) \\&= \frac{1}{2} e^{i\frac{\varphi_0+\varphi_1}{2}} \left(2i \sin \frac{\varphi_0 - \varphi_1}{2} \right) \\&= -ie^{i\frac{\varphi_0+\varphi_1}{2}} \sin \frac{\varphi_1 - \varphi_0}{2}\end{aligned}$$

resorting to $x = \frac{x+y}{2} + \frac{x-y}{2}$
and Euler formula $e^{i\varphi} = \cos \varphi + i \sin \varphi$

What is the probability of an atom going from $|0\rangle$ to $|1\rangle$?

Then the probability:

$$\begin{aligned} P_{10} &= |A_{10}|^2 \\ &= \left| -ie^{i\frac{\varphi_0 + \varphi_1}{2}} \sin \frac{\varphi_1 - \varphi_0}{2} \right|^2 \\ &= \left| \sin \frac{\varphi_1 - \varphi_0}{2} \right|^2 \\ &= \frac{1}{2} - \underbrace{\frac{1}{2} \cos(\varphi_1 - \varphi_0)}_{\text{interference}} \end{aligned}$$

resorting to $\cos 2\theta = 1 - 2\sin^2 \theta$

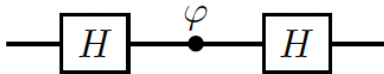
A bulk calculation ...

The results of these calculations for all pairs begin-end states can be expressed as follows:

- the effect of each interaction is described by a **matrix of transition amplitudes**, and
- one resorts to **matrix multiplication** to compose the sequence of independent interactions.

$$\begin{aligned}
 A &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} e^{i\varphi_0} & 0 \\ 0 & e^{i\varphi_1} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \\
 &= e^{i\frac{\varphi_0 + \varphi_1}{2}} \begin{bmatrix} \cos \frac{\varphi}{2} & -i \sin \frac{\varphi}{2} \\ -i \sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \end{bmatrix} \quad \text{making } \varphi = \varphi_1 - \varphi_0 \\
 &\equiv \begin{bmatrix} \cos \frac{\varphi}{2} & -i \sin \frac{\varphi}{2} \\ -i \sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \end{bmatrix} \quad \text{ignoring global phase, to discuss later ...}
 \end{aligned}$$

My first quantum circuit



- A **wire** represents a two-dimensional **state** (a **qubit**)
- Three **gates** describing quantum **operations**:

$$\underbrace{H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix}}_{\text{Hadamard gate}} \quad \text{and} \quad \underbrace{P_{\varphi} = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{bmatrix}}_{\text{Phase shift gate}}$$

A simple matrix multiplication yields

$$A = HP_{\varphi}H = \begin{bmatrix} \cos \frac{\varphi}{2} & -i \sin \frac{\varphi}{2} \\ -i \sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \end{bmatrix} = \begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix}$$

My first quantum circuit

which, expressed in a functional way, gives

$$A|0\rangle = \cos \frac{\varphi}{2} |0\rangle + -i \sin \frac{\varphi}{2} |1\rangle$$

$$A|1\rangle = -i \sin \frac{\varphi}{2} |0\rangle + \cos \frac{\varphi}{2} |1\rangle$$

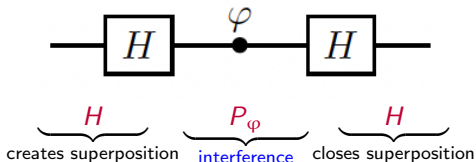
as read from

$$A = HP_{\varphi}H = \begin{bmatrix} \cos \frac{\varphi}{2} & -i \sin \frac{\varphi}{2} \\ -i \sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \end{bmatrix}$$

Clearly, for $\varphi = 0$, i.e. in the absence of any phase shift, $A|0\rangle = |0\rangle$ and $A|1\rangle = |1\rangle$, leading to conclude that

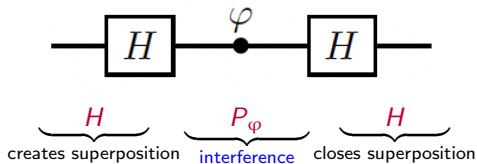
$$HH = Id$$

The golden pattern



- H creates/closes a **uniform superposition**: it is the source of a natural parallelism,
- but the **crucial** role in controlling interference is located in P_φ .

Lessons learnt for what follows



generalizes to a basic **recipe** in quantum algorithms (to be discussed).

What is quantum computation?

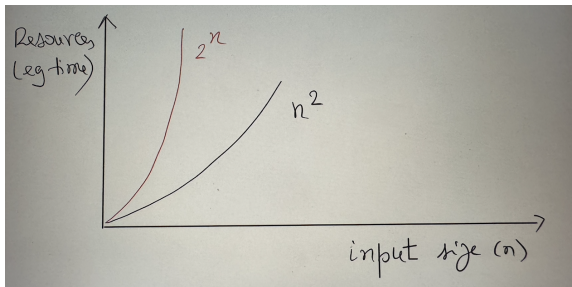
A complex multi-particle quantum **interference involving many computational paths** through a computing device.

Its challenge is **to shape quantum interference** through a sequence of computational steps, enhancing probabilities of the **correct** outputs and suppressing probabilities of the **wrong** ones.

Superposition and interference what for?

Quantum computing has nothing to say on the boundaries of **computability**, but reduces the **complexity** of algorithms to address several problems.

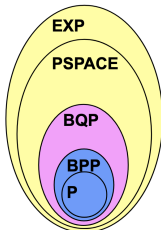
The limits of practicality



Paying a visit to old acquaintances

- EXP** Problems requiring exponential time to be solved
- PSPACE** Problems requiring polynomial space to be solved
- BQP** Problems solved in quantum machines in bounded error polynomial time
- BPP** Problems solved in probabilistic machines in bounded error polynomial time
- P** Problems solved in deterministic classical machines in polynomial time

$$P \subseteq BPP \subseteq BQP \subseteq PSPACE \subseteq EXP$$

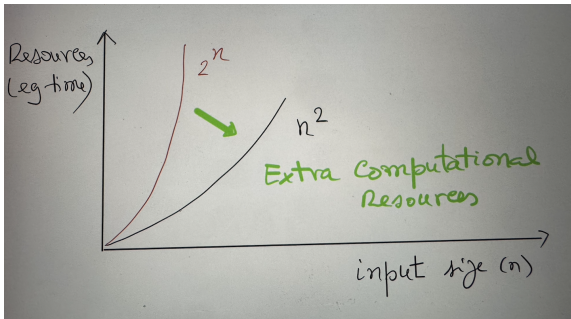


Superposition and interference as computational resources

A problem does not become **tractable** by increasing computational power:
more of the same keeps similar circumstances.

New ideas & new resources make a difference

The key: Quantum effects as computational resources



Decoherence

The laws of quantum computation just described assume a **perfectly isolated system**.

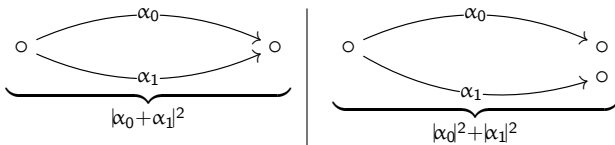
- In practice, any complex quantum system cannot avoid some spurious **interactions with the environment**.
- Thus, when computing the system's evolution, one must take into account not only its internal configurations but also those of its environment.

This is known as **decoherence** and explains why

- we hardly see quantum interference on a daily basis
- and isolation is a main technological challenge in the design/construction of quantum computing devices.

Two scenarios

- **Perfect isolation:** the environment does not hold any physical record of the path taken to produce an output.
- **Total decoherence:** the system produces an output but the environment has a record of the path taken: there are no alternative ways to reach that output, but a single path to each of the following situations (joint states):
 1. The output was produced, but the environment knows the path labelled by α_0 was taken,
 2. or similarly for α_1 .



Two scenarios

- **Perfect isolation**: amplitudes are added and generate a probability for the output to be produced, taking both paths in superposition
- **Total decoherence**: probabilities are computed for each path and summed as such — **interference** provided by superposition is **lost**.

The general scenario

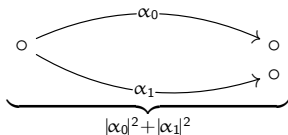
Interference is limited by a **visibility** parameter — v — ranging from

- **0**: total decoherence; no interference
- **1**: no decoherence; full interference

The effect of decoherence

$$p_0 + p_1 + 2\mathfrak{v}\sqrt{p_0p_1}\cos(\phi_1 - \phi_0)$$

Parameter $\mathfrak{v}$ quantifies the **degree of distinguishability between the disrupted outputs**: *the more the environment knows the less interference is observed more classic becomes the overall computation.*



Fact

Decoherence destroys (quantum) interference.